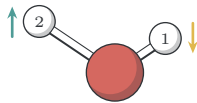


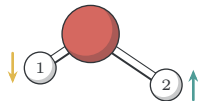
$\mathbf{X}$

$$\xrightarrow[\chi_{\text{stat}}(\sigma) = -1]{\sigma = (12)}$$



$\sigma \mathbf{X} = \mathbf{X} \mathbf{P}_\sigma$

$$\xrightarrow[\chi_{\pm}(E^*) = \pm 1]{E^*}$$



$-\mathbf{X} \mathbf{P}_\sigma = E^* \sigma \mathbf{X}$

$$\Psi \begin{bmatrix} \uparrow & \downarrow \\ 1 & 2 \end{bmatrix} \Psi(\mathbf{X})$$



$$-\begin{bmatrix} \downarrow & \uparrow \\ 1 & 2 \end{bmatrix}$$

$$\Psi(\mathbf{X} \mathbf{P}_\sigma) = \chi_{\text{stat}}(\sigma) \sigma \cdot \Psi(\mathbf{X})$$



$$\mp \begin{bmatrix} \downarrow & \uparrow \\ 1 & 2 \end{bmatrix}$$

$$\Psi(-\mathbf{X} \mathbf{P}_\sigma) = \chi_{\pm} \Psi(\mathbf{X} \mathbf{P}_\sigma)$$